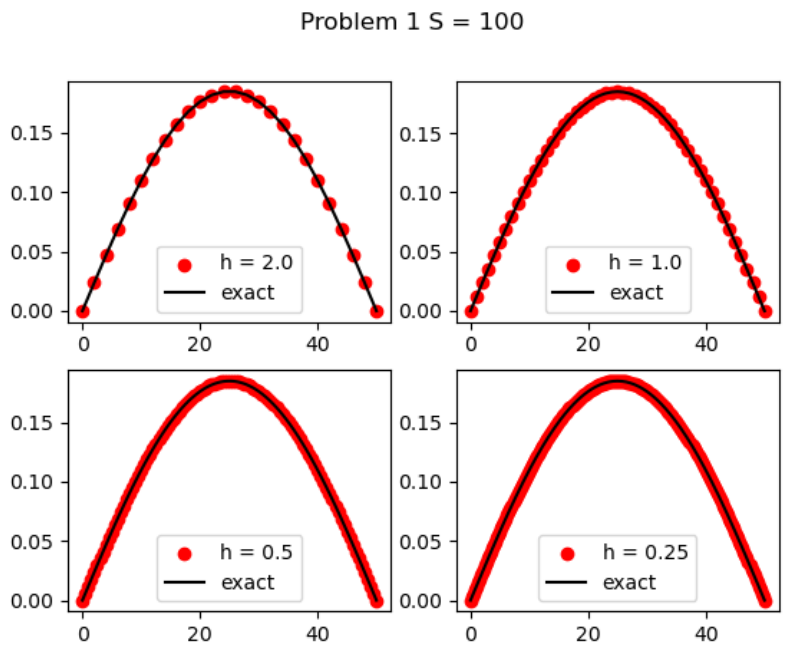
MATH 610 PA 1 Report

Jordan Hoffart

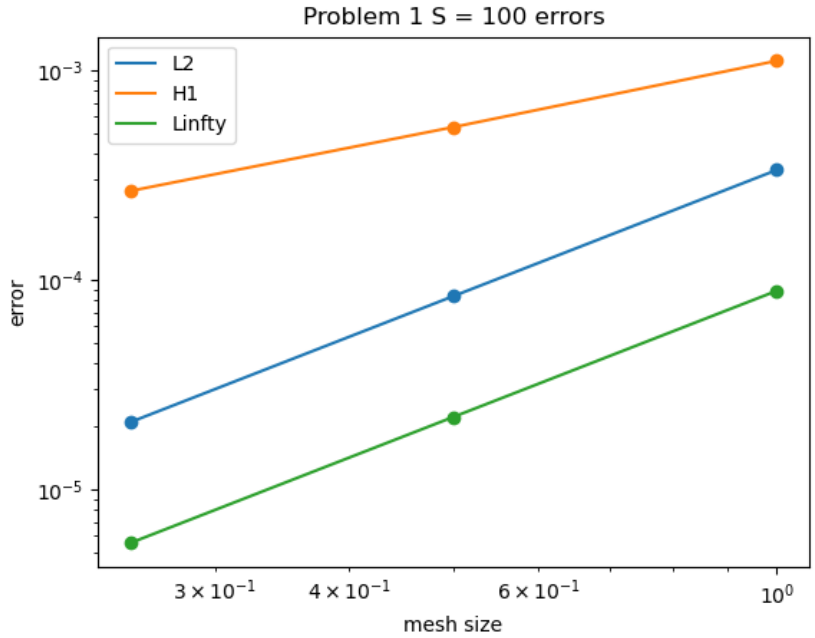
March 4, 2024

Problem 1

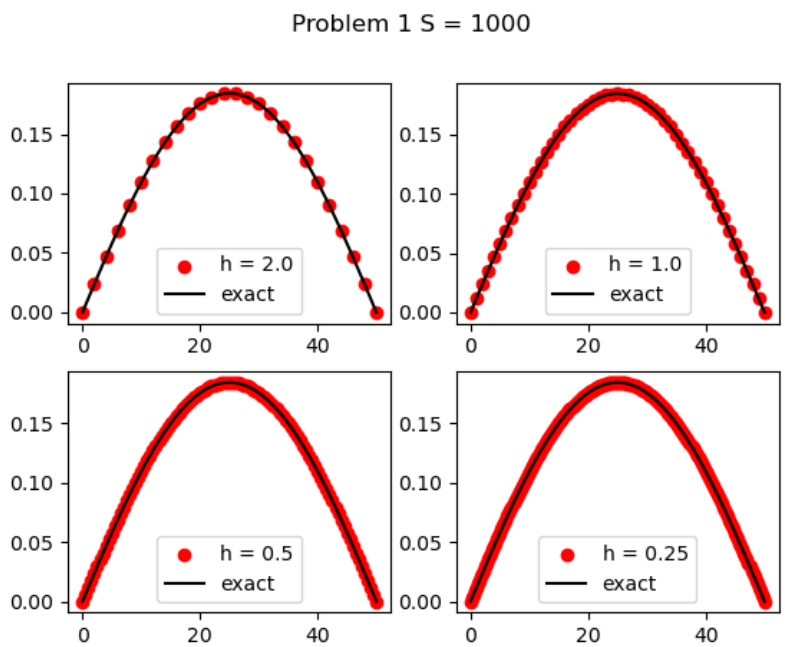
$S = 100$



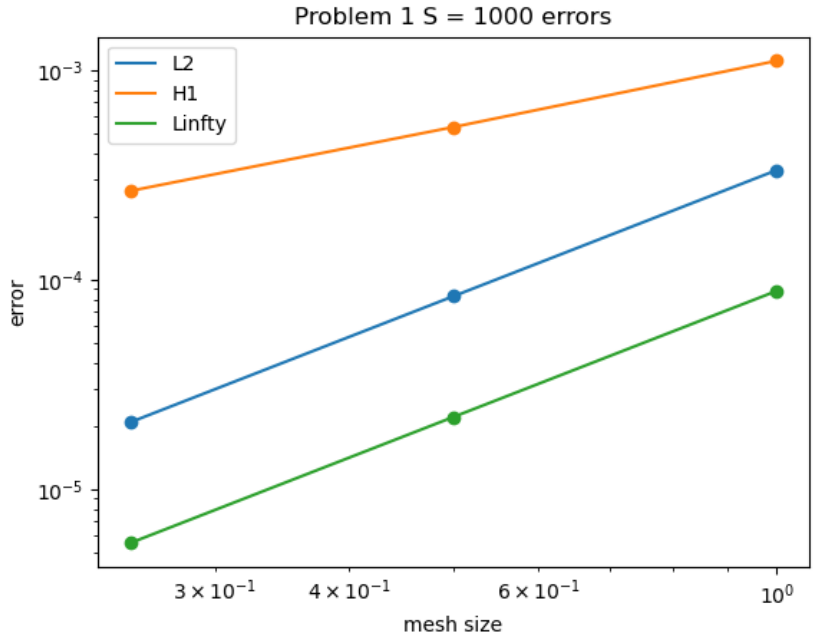
mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	1.34E-03	2.50E-03	3.55E-04	0.00E+00	0.00E+00	0.00E+00
1.00E+00	3.35E-04	1.11E-03	8.87E-05	2.00E+00	1.17E+00	2.00E+00
5.00E-01	8.37E-05	5.36E-04	2.22E-05	2.00E+00	1.05E+00	2.00E+00
2.50E-01	2.09E-05	2.65E-04	5.55E-06	2.00E+00	1.01E+00	2.00E+00


Problem 1 S = 100 errors
L2
H1
Linfty
10^{-3}
10^{-4}
10^{-5}
error
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}
mesh size

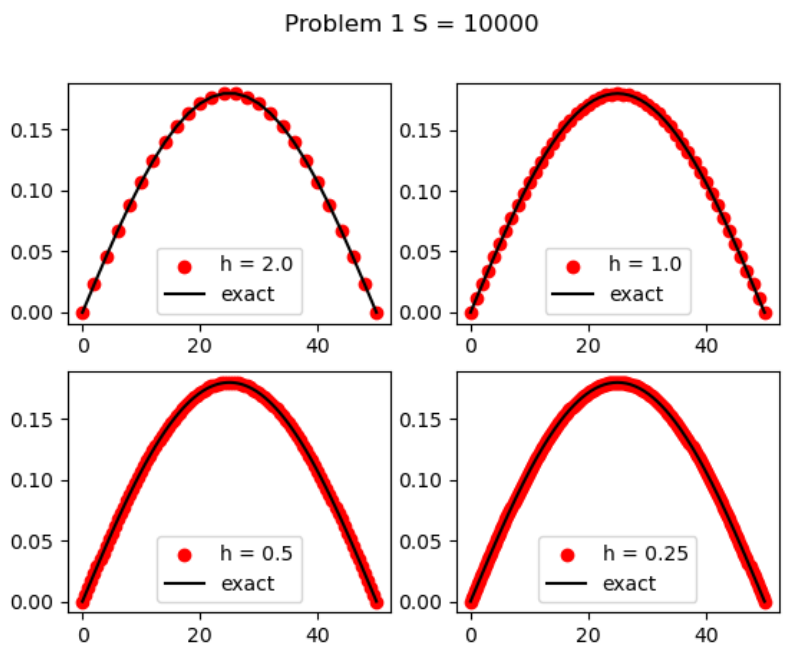
$S = 1000$



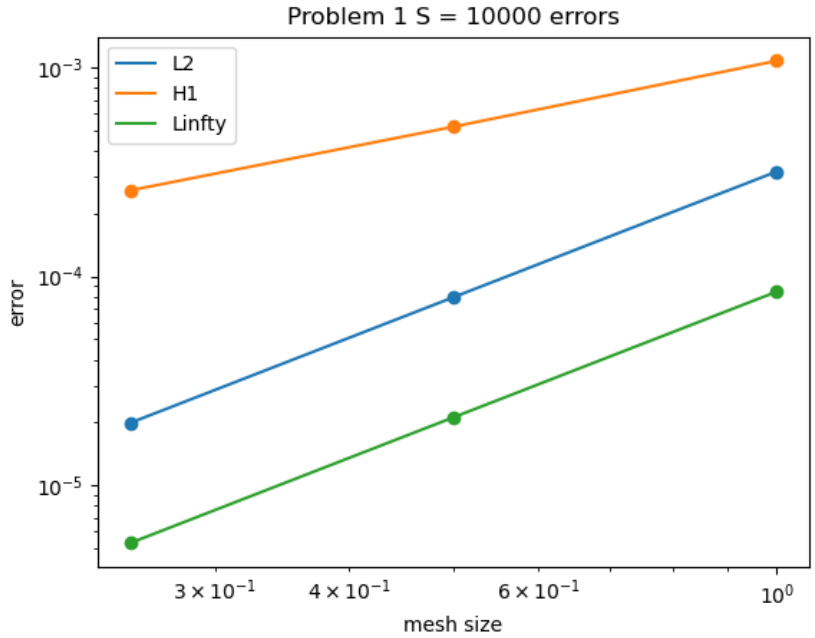
mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	1.33E-03	2.50E-03	3.53E-04	0.00E+00	0.00E+00	0.00E+00
1.00E+00	3.33E-04	1.11E-03	8.83E-05	2.00E+00	1.17E+00	2.00E+00
5.00E-01	8.33E-05	5.34E-04	2.21E-05	2.00E+00	1.05E+00	2.00E+00
2.50E-01	2.08E-05	2.65E-04	5.52E-06	2.00E+00	1.01E+00	2.00E+00


Problem 1 S = 1000 errors
L2
H1
Linfty
10^{-3}
10^{-4}
10^{-5}
error
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}
mesh size

$S = 10000$

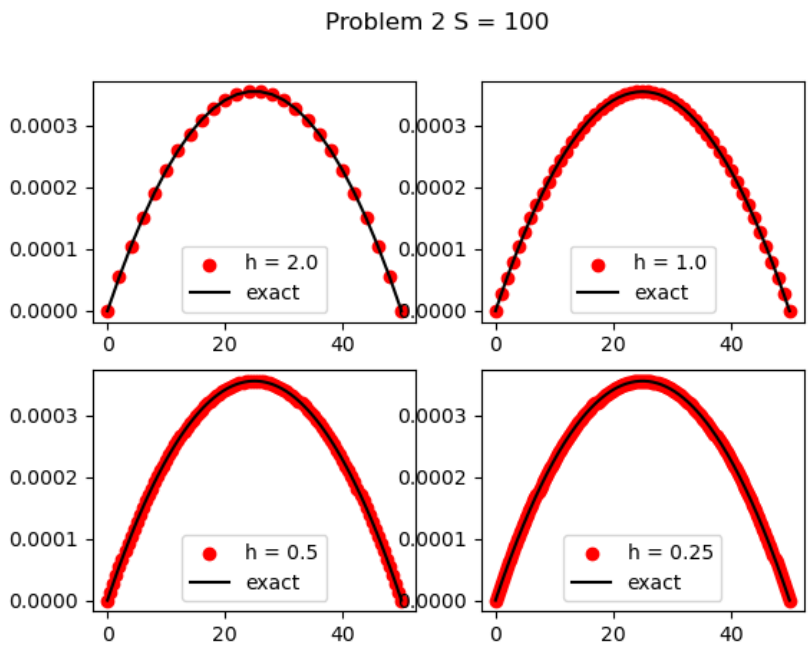


mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	1.27E-03	2.42E-03	3.38E-04	0.00E+00	0.00E+00	0.00E+00
1.00E+00	3.18E-04	1.08E-03	8.45E-05	2.00E+00	1.17E+00	2.00E+00
5.00E-01	7.95E-05	5.21E-04	2.11E-05	2.00E+00	1.05E+00	2.00E+00
2.50E-01	1.99E-05	2.58E-04	5.29E-06	2.00E+00	1.01E+00	2.00E+00

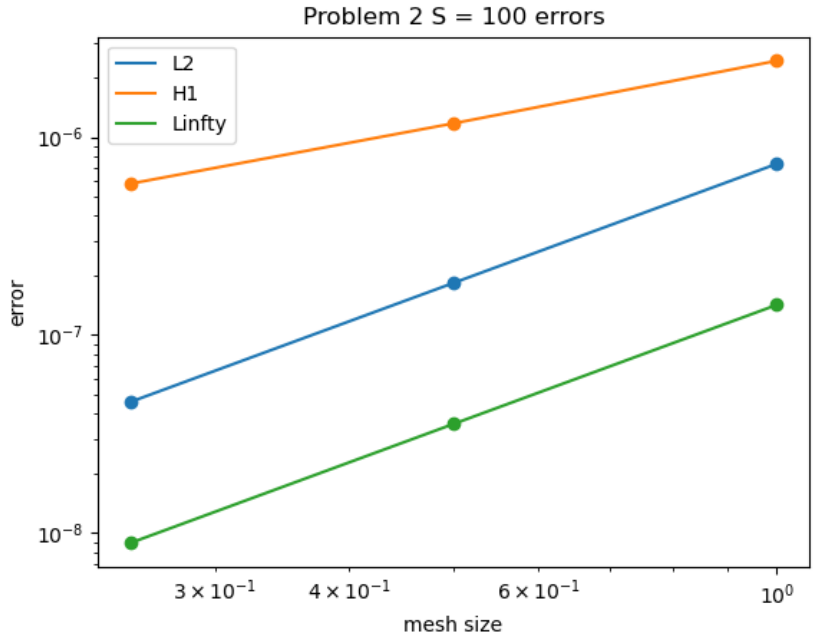

Problem 1 S = 10000 errors
L2
H1
Linfty
error
mesh size
10^{-3}
10^{-4}
10^{-5}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

Problem 2

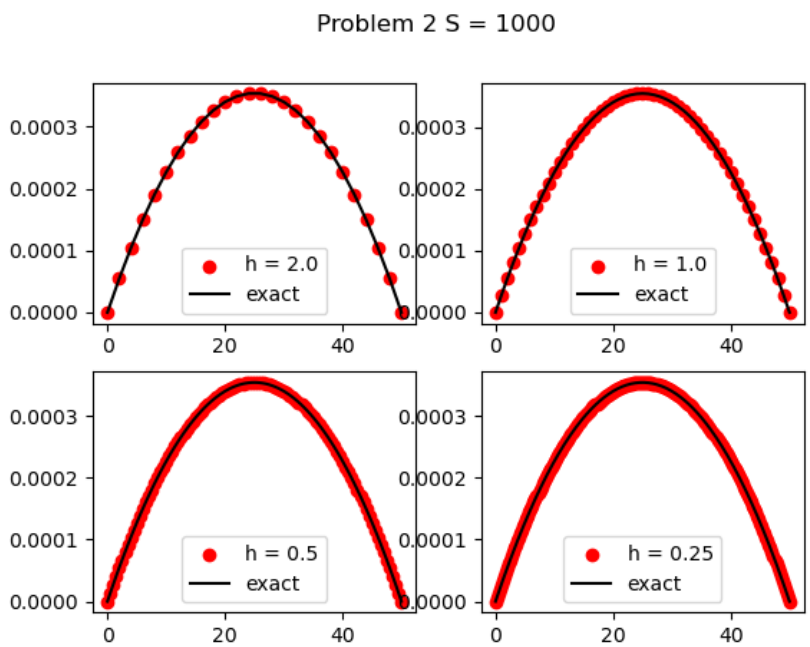
$S = 100$



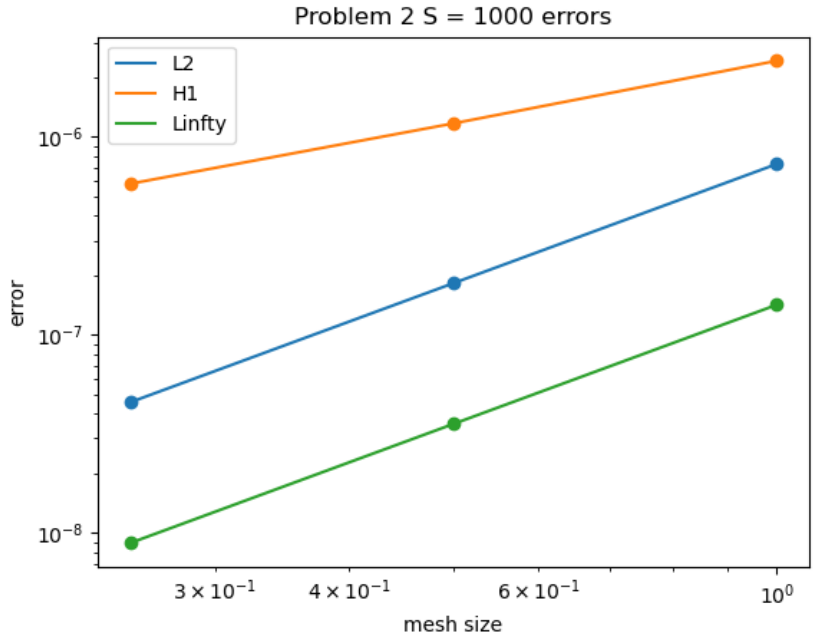
mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	2.93E-06	5.49E-06	5.68E-07	0.00E+00	0.00E+00	0.00E+00
1.00E+00	7.33E-07	2.43E-06	1.42E-07	2.00E+00	1.17E+00	2.00E+00
5.00E-01	1.83E-07	1.17E-06	3.55E-08	2.00E+00	1.05E+00	2.00E+00
2.50E-01	4.58E-08	5.82E-07	8.88E-09	2.00E+00	1.01E+00	2.00E+00


Problem 2 S = 100 errors
L2
H1
Linfty
error
mesh size
10^{-6}
10^{-7}
10^{-8}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

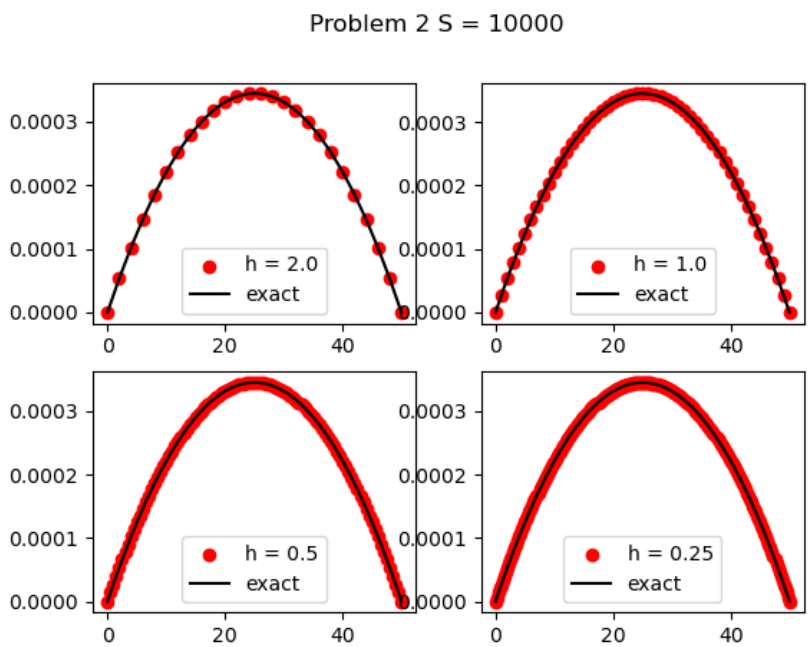
$S = 1000$



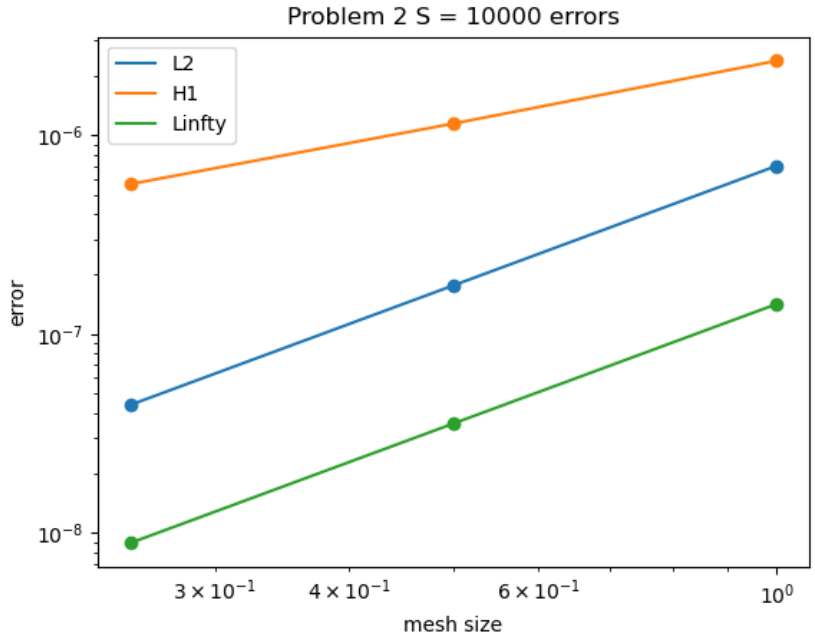
mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	2.92E-06	5.47E-06	5.68E-07	0.00E+00	0.00E+00	0.00E+00
1.00E+00	7.30E-07	2.43E-06	1.42E-07	2.00E+00	1.17E+00	2.00E+00
5.00E-01	1.83E-07	1.17E-06	3.55E-08	2.00E+00	1.05E+00	2.00E+00
2.50E-01	4.56E-08	5.80E-07	8.88E-09	2.00E+00	1.01E+00	2.00E+00


Problem 2 S = 1000 errors
L2
H1
Linfty
error
mesh size
10^{-6}
10^{-7}
10^{-8}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

$S = 10000$



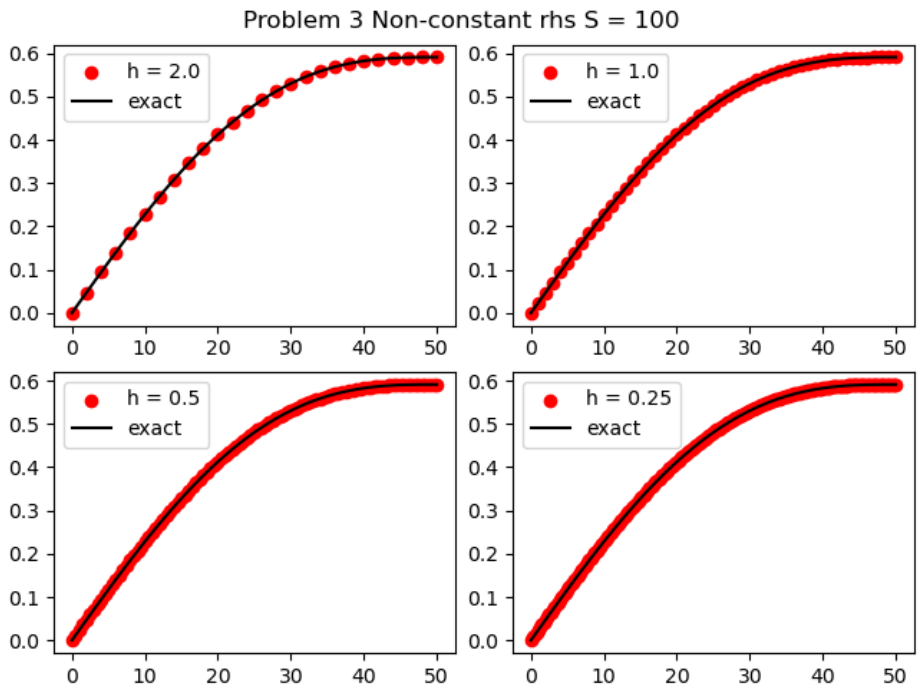
mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	2.81E-06	5.33E-06	5.66E-07	0.00E+00	0.00E+00	0.00E+00
1.00E+00	7.03E-07	2.37E-06	1.42E-07	2.00E+00	1.17E+00	2.00E+00
5.00E-01	1.76E-07	1.15E-06	3.55E-08	2.00E+00	1.05E+00	2.00E+00
2.50E-01	4.39E-08	5.68E-07	8.87E-09	2.00E+00	1.01E+00	2.00E+00


Problem 2 S = 10000 errors
L2
H1
Linfty
error
mesh size
10^{-6}
10^{-7}
10^{-8}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

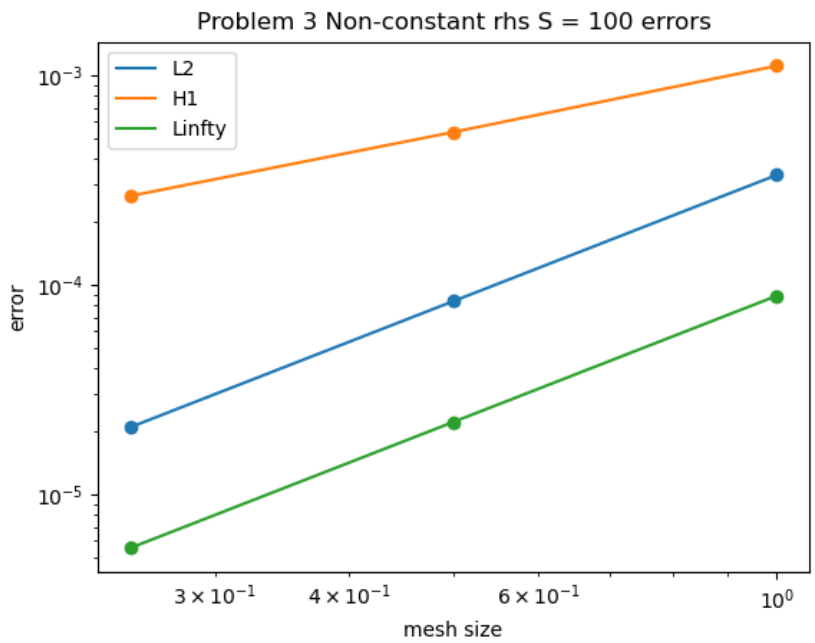
Problem 3

Non-constant rhs

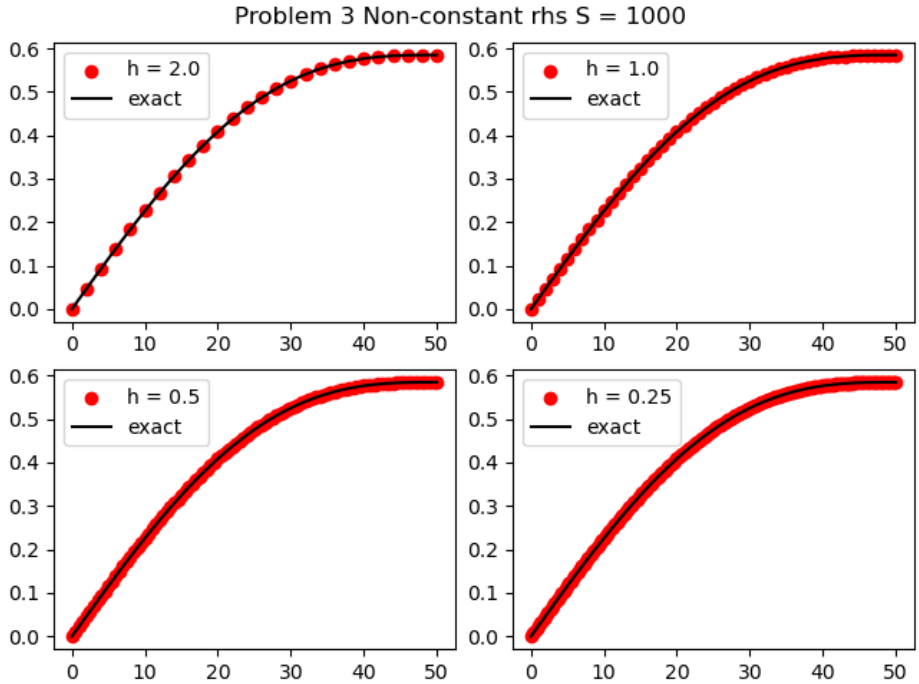
$S = 100$



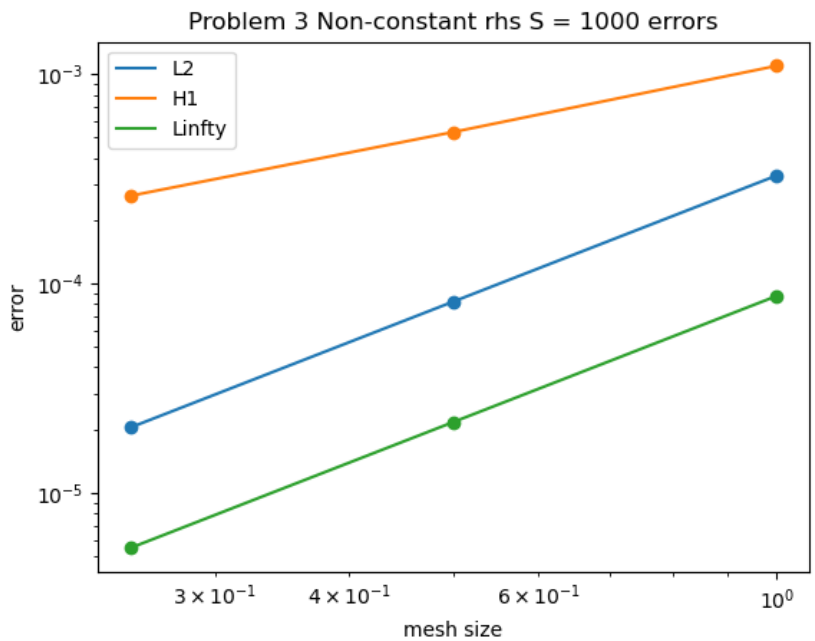
mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	1.34E-03	2.50E-03	3.55E-04	0.00E+00	0.00E+00	0.00E+00
1.00E+00	3.34E-04	1.11E-03	8.86E-05	2.00E+00	1.17E+00	2.00E+00
5.00E-01	8.36E-05	5.35E-04	2.22E-05	2.00E+00	1.05E+00	2.00E+00
2.50E-01	2.09E-05	2.65E-04	5.54E-06	2.00E+00	1.01E+00	2.00E+00


Problem 3 Non-constant rhs S = 100 errors
L2
H1
Linfty
error
mesh size
10^{-3}
10^{-4}
10^{-5}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

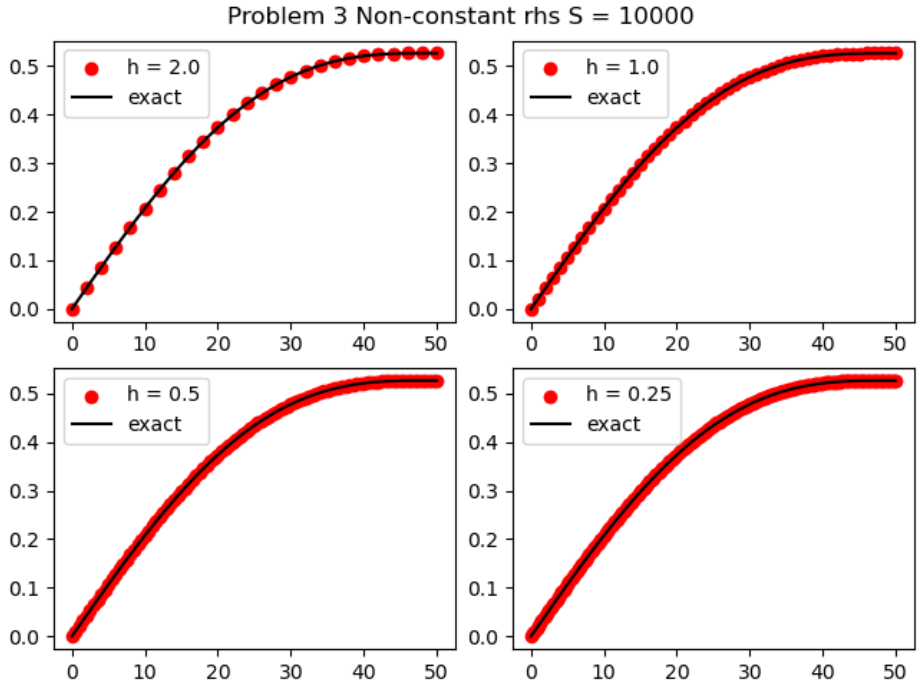
$S = 1000$



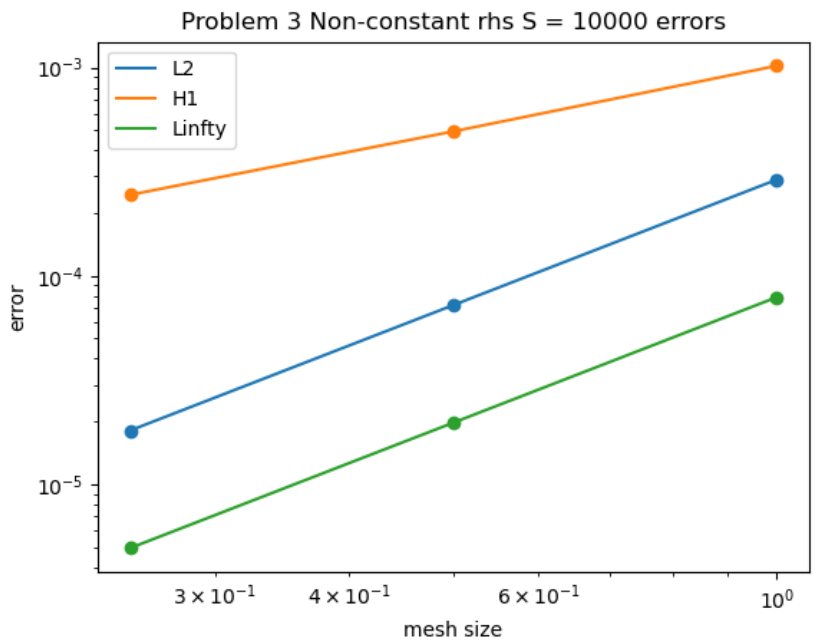
mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	1.32E-03	2.48E-03	3.51E-04	0.00E+00	0.00E+00	0.00E+00
1.00E+00	3.29E-04	1.10E-03	8.76E-05	2.00E+00	1.17E+00	2.00E+00
5.00E-01	8.24E-05	5.31E-04	2.19E-05	2.00E+00	1.05E+00	2.00E+00
2.50E-01	2.06E-05	2.63E-04	5.48E-06	2.00E+00	1.01E+00	2.00E+00


Problem 3 Non-constant rhs S = 1000 errors
L2
H1
Linfty
error
mesh size
10^{-3}
10^{-4}
10^{-5}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

$S = 10000$

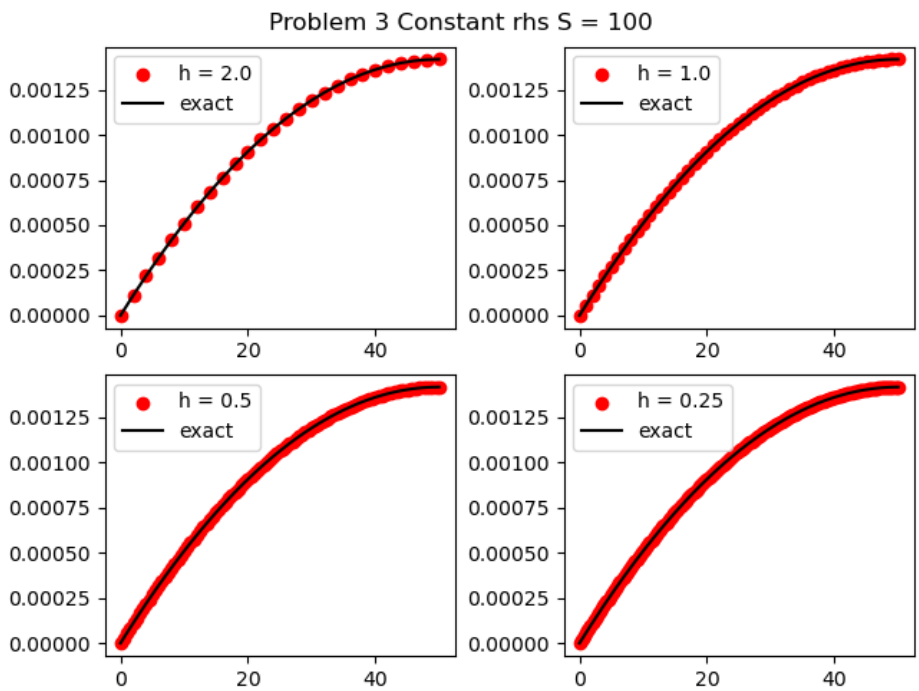


mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	1.16E-03	2.27E-03	3.16E-04	0.00E+00	0.00E+00	0.00E+00
1.00E+00	2.90E-04	1.02E-03	7.90E-05	2.00E+00	1.16E+00	2.00E+00
5.00E-01	7.24E-05	4.93E-04	1.98E-05	2.00E+00	1.05E+00	2.00E+00
2.50E-01	1.81E-05	2.45E-04	4.94E-06	2.00E+00	1.01E+00	2.00E+00

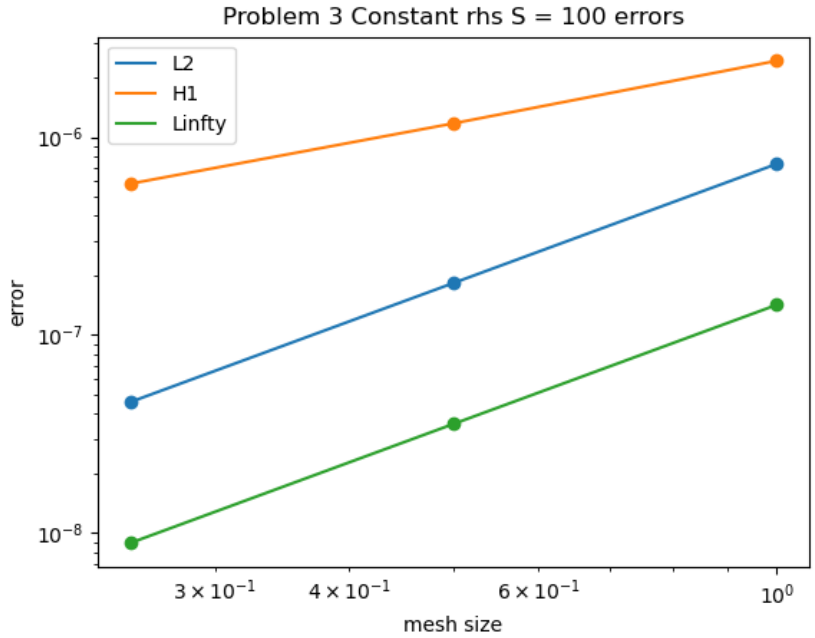

Problem 3 Non-constant rhs S = 10000 errors
L2
H1
Linfty
error
mesh size
10^{-3}
10^{-4}
10^{-5}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

Constant rhs

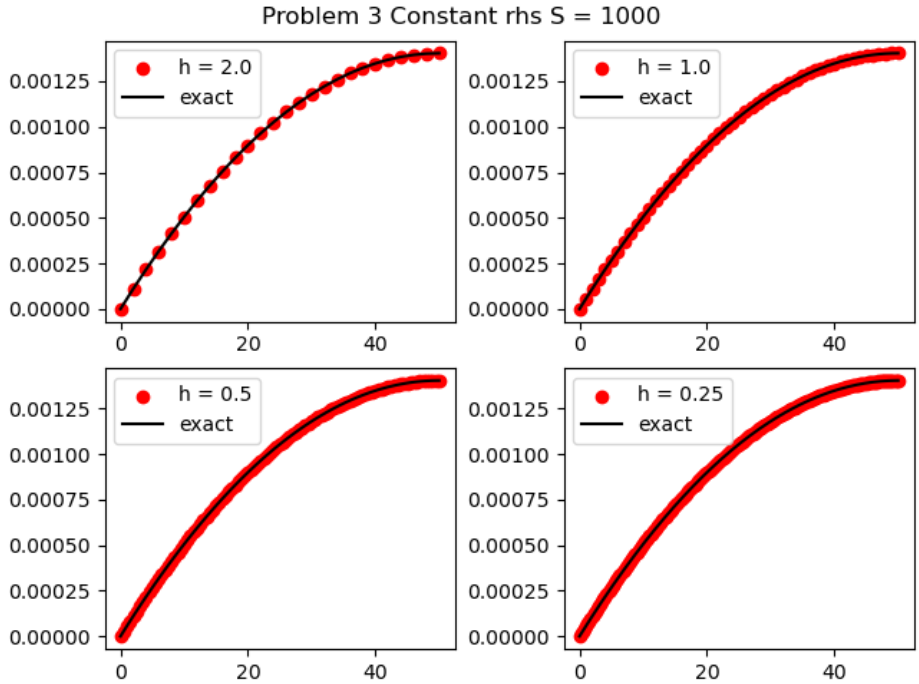
$S = 100$



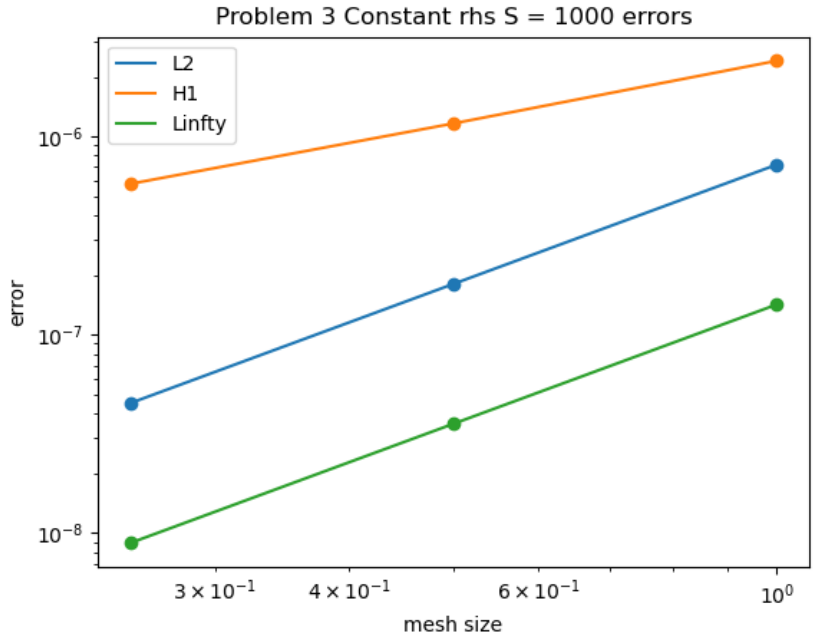
mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	2.93E-06	5.48E-06	5.68E-07	0.00E+00	0.00E+00	0.00E+00
1.00E+00	7.32E-07	2.43E-06	1.42E-07	2.00E+00	1.17E+00	2.00E+00
5.00E-01	1.83E-07	1.17E-06	3.55E-08	2.00E+00	1.05E+00	2.00E+00
2.50E-01	4.58E-08	5.81E-07	8.88E-09	2.00E+00	1.01E+00	2.00E+00


Problem 3 Constant rhs S = 100 errors
L2
H1
Linfty
error
mesh size
10^{-6}
10^{-7}
10^{-8}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

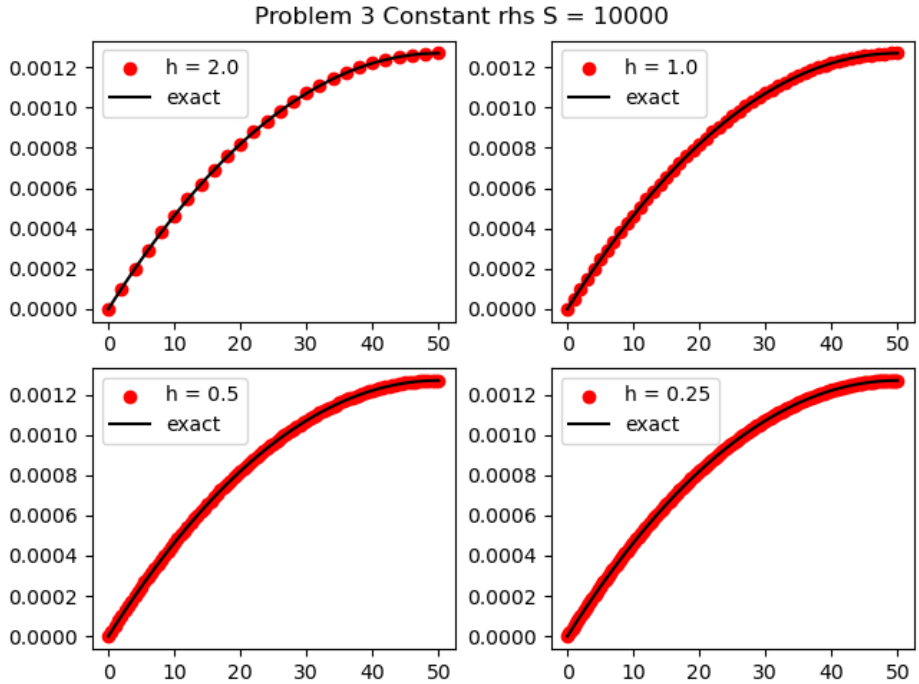
$S = 1000$



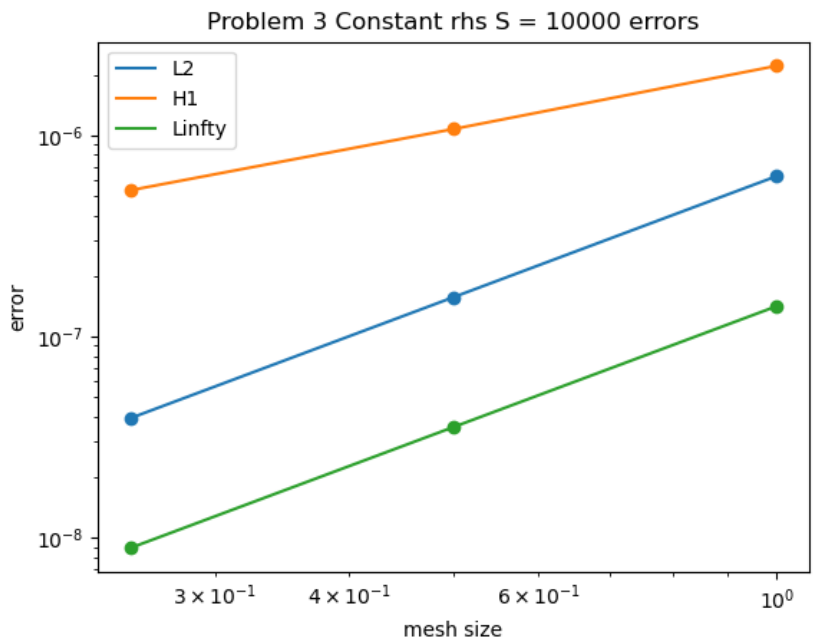
mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	2.88E-06	5.43E-06	5.68E-07	0.00E+00	0.00E+00	0.00E+00
1.00E+00	7.21E-07	2.41E-06	1.42E-07	2.00E+00	1.17E+00	2.00E+00
5.00E-01	1.80E-07	1.16E-06	3.55E-08	2.00E+00	1.05E+00	2.00E+00
2.50E-01	4.51E-08	5.76E-07	8.88E-09	2.00E+00	1.01E+00	2.00E+00


Problem 3 Constant rhs S = 1000 errors
L2
H1
Linfty
error
mesh size
10^{-6}
10^{-7}
10^{-8}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

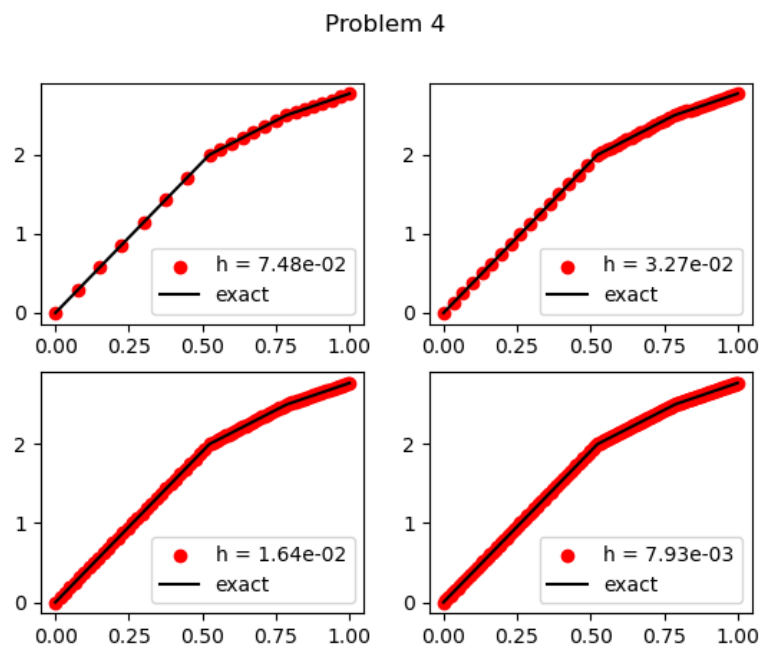
$S = 10000$



mesh size	L2 error	H1 error	Linf error	L2 rate	H1 rate	Linf rate
2.00E+00	2.51E-06	4.93E-06	5.64E-07	0.00E+00	0.00E+00	0.00E+00
1.00E+00	6.27E-07	2.21E-06	1.41E-07	2.00E+00	1.16E+00	1.99E+00
5.00E-01	1.57E-07	1.07E-06	3.54E-08	2.00E+00	1.04E+00	2.00E+00
2.50E-01	3.92E-08	5.32E-07	8.87E-09	2.00E+00	1.01E+00	2.00E+00


Problem 3 Constant rhs S = 10000 errors
L2
H1
Linfty
error
mesh size
10^{-6}
10^{-7}
10^{-8}
3×10^{-1}
4×10^{-1}
6×10^{-1}
10^{0}

Problem 4



mesh size	L2 error	H1 error	Linf error
7.48E-02	9.85E-15	2.11E-14	1.79E-14
3.27E-02	1.13E-14	3.14E-14	2.19E-14
1.64E-02	2.35E-14	9.34E-14	3.09E-14
7.93E-03	4.21E-13	9.86E-13	8.29E-13